From a Go converter to a composed architecture document.

A library caller supplies HTML, chooses the document contract, and receives deterministic PDF bytes without shelling out to a browser or a system binary.

Core idea. The public API is a narrow ownership boundary around settings, page objects, conversion context, and output bytes. Internal layout and PDF stages remain replaceable behind that boundary.

server-generated HTML | context-aware | PDF bytes in memory

1 · Configure

Global page geometry, margins, background, and local-file policy.

→

2 · Add object

Attach one or more HTML pages with object-level overrides.

→

3 · Convert

Thread context through load, parse, style, layout, paint, and write.

→

4 · Consume

Read stable PDF bytes and store or stream them from the host service.

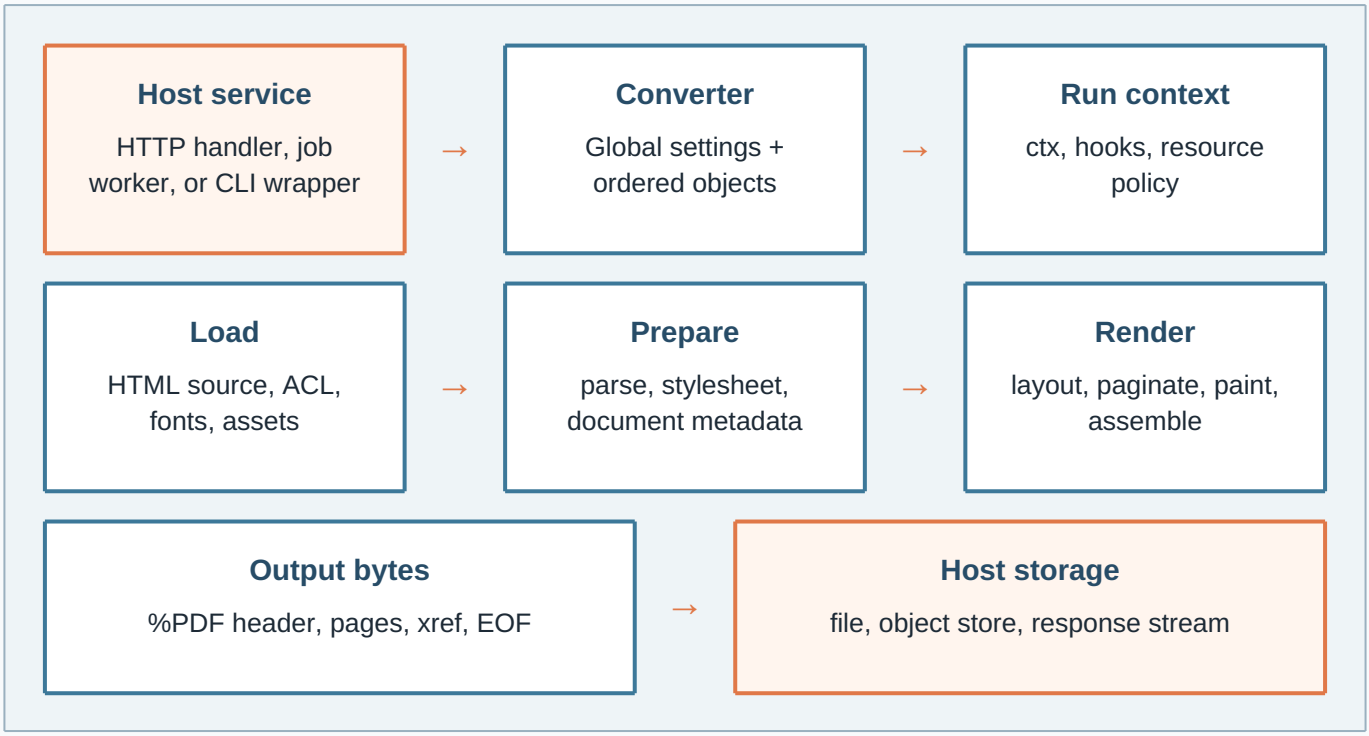
5
architecture pages in this example

0
external browser processes

1
explicit output ownership boundary

The request path is explicit at every seam.

The converter owns configuration and output. Each page object owns its source and local overrides. The context gives callers a cancellation and deadline boundary.



Ownership rules

- `AddObject` copies public settings before conversion.
- `Output` returns a copy of the completed bytes.
- A converter is used by one conversion flow at a time.
- Failures return as errors before partial output is published.

Signals exposed to callers

- Context cancellation and deadlines.
- Phase names and integer progress callbacks.
- Info, warning, and error conversion hooks.
- Typed sentinel errors for invalid boundaries.

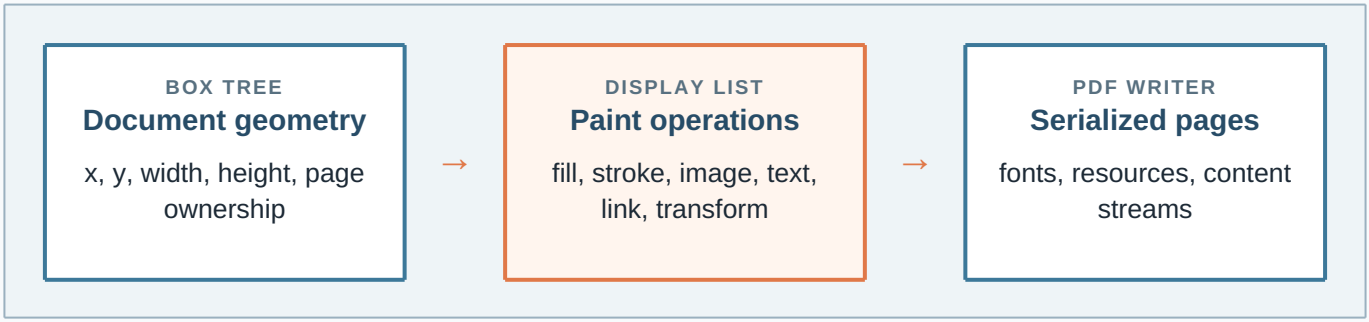
Design note. A library caller never needs to know which internal package performs a stage. The public contract stays useful even as layout and PDF implementation details evolve.

Rendering is a staged document pipeline.

The renderer turns one source document into an ordered display model. Each stage contributes a stronger invariant and keeps its work visible to the next stage.

Stage	Input → output	Invariant carried forward
Load	URL/path/inline bytes → source tree	ACL checks and resource policy are applied before fetches.
Parse	HTML/CSS text → nodes and rules	Malformed but recoverable markup remains traversable.
Style	rules + ancestry → resolved styles	Inherited values, custom properties, and media are resolved once.
Layout	styled tree → boxes and flow positions	Widths, heights, text runs, tables, flex, and grid have geometry.
Paint	boxes → display operations	Operations preserve order, links, backgrounds, and text metadata.
Write	operations → PDF bytes	Pages, resources, xref, and EOF form a readable PDF artifact.

Display-list handoff



Why a display list?

Pagination can move a group of operations without rebuilding source nodes. Tests can inspect paint order and page ownership before serialization.

Why keep the API above it?

Callers need a document result, not a renderer vocabulary. The public surface can remain small while internal contracts stay testable.

Safe defaults make the library embeddable.

A service embedding the converter must be able to decide which inputs can read local files, which resources can be fetched, and when a request should stop.

ACL

Local resources

Local-file access is blocked by default. The global enable flag and object-level unblock form an explicit pair.

CTX

Cancellation

The caller supplies a context. Load, conversion, and output stages can stop without leaving a published partial PDF.

LIMITS

Resource bounds

Object, copy, page, stylesheet, and source limits prevent accidental amplification from untrusted inputs.

```
converter:= gowkhtmltopdf.NewConverter()
converter.Global().Set("enablelocalfileaccess", "true")
page:= gowkhtmltopdf.NewObjectSettings().SetPage(inputPath)
page.Set("load.blocklocalfileaccess", "false")
converter.AddObject(page)
err:= converter.Convert(ctx)
```

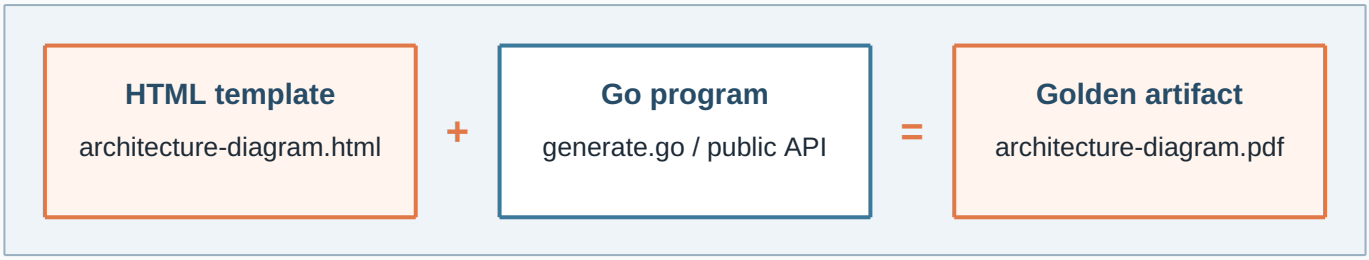
Failure boundaries

Condition	Observed by host	Recommended host action
Missing page object	typed no-page error	return a 4xx configuration response
Blocked local file	load/resource error	keep ACL blocked and report the source
Context cancellation	context error	stop work and do not publish output
Invalid setting	setting validation error	reject the request before conversion

internal stage
caller decision
published contract

One small example can carry a complete document contract.

The generator beside this template is intentionally ordinary Go: configure, add a local page, convert with a context, validate the page count, and write the bytes.



Integration checklist

1. Choose whether input is a path or in-memory body.
2. Set page geometry and rendering policy explicitly.
3. Pair local-file settings when local resources are intended.
4. Pass the host request context to `Convert`.
5. Write or stream `Output` only after success.

What this artifact demonstrates

- Five stable, explicit document pages.
- Architecture diagrams built from print-safe CSS boxes.
- Text, tables, code, metadata, and page breaks.
- A reproducible output path beside its source template.

Closing contract. The HTML remains understandable as a document, the Go program remains understandable as an integration example, and the PDF is a reviewable artifact produced by the same public API that an embedding service would call.

HTML in · Go API · PDF out